Stochastic Programming

Chapter 3: Basic Properties of Stochastic Programming problems

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Slides based on book: J.R. Birge, F. Louveaux, *Introduction to Stochastic Programming*, Springer, 1997.

Contents

Outline

Original formulation

- Remember the classical two-stage stochastic linear problem with fixed recourse

$$\begin{array}{ll} \min\limits_{x,y(\omega)} & c^\top x + E_\xi[\min \;\; q^\top(\omega)y(\omega)] \\ \text{s.to} & Ax = b \\ & T(\omega)x + Wy(\omega) = h(\omega) \\ & x \geq 0, y(\omega) \geq 0 \end{array}$$

$$\begin{array}{l} x \in \mathbb{R}^{n_1} \\ c \in \mathbb{R}^{n_1} \\ b \in \mathbb{R}^{m_1} \\ A \in \mathbb{R}^{m_1 \times n_1} \end{array} \qquad \begin{array}{l} y(\omega) \in \mathbb{R}^{n_2} \\ q(\omega) \in \mathbb{R}^{n_2} \\ h(\omega) \in \mathbb{R}^{m_2} \\ T(\omega) \in \mathbb{R}^{m_2 \times n_1} \\ W \in \mathbb{R}^{m_2 \times n_2} \end{array}$$

$$\xi^\top(\omega) = \left(q^\top(\omega), h^\top(\omega), T_1(\omega), \ldots, T_{m_2}(\omega)\right) \in \mathbb{R}^{n_2+m_2+m_2\times n_1 = N}$$

- Let $\Xi \subseteq \mathbb{R}^N$ be the support of ξ, i.e., the smallest closed subset of $\mathbb{R}^N$ such that $Pr(\xi \in \Xi) = 1$.

Deterministic Equivalent Problem formulation

- Consider the DEP formulation

$$\begin{aligned} \min \quad & c^\top x + \mathcal{Q}(x) \\ \text{s. to} \quad & Ax = b \\ & x \geq 0 \end{aligned}$$

$$\begin{aligned} Q(x,\omega) = \quad & \min_y q^\top(\omega) y \\ \text{s. to} \quad & Wy = h(\omega) - T(\omega)x \\ & y \geq 0 \end{aligned}$$

where $\mathcal{Q}(x) = E_\xi[Q(x,\omega)]$

- Given x, we compute $Q(x,\omega)$ via minimization. We will denote
  - $Q(x,\omega) = -\infty$ if problem is unbounded
  - $Q(x,\omega) = \infty$ if problem is infeasible
- Since the recourse function $\mathcal{Q}(x)$ is an expectation, we will consider separately when $\xi(\omega)$ is discrete or continuous random variable.

Case $\xi(\omega)$ discrete

- If $\xi(\omega)$ follows a finite discrete distribution, then

$$\mathcal{Q}(x) = \sum_{s \in S} p_s Q(x, \omega_s)$$

- Let's suppose $Q(x, \omega_s) = \infty$, $Q(x, \omega_r) = -\infty$ for some ω_s, $\omega_r \in \Omega$. Here, we make the convention $+\infty - \infty = +\infty$.

- In these cases, we'll consider that x give us to a infeasible second stage (independent to the existence of another ω_r such that $Q(x, \omega_r) = -\infty$.

Case $\xi(\omega)$ discrete

Let's define the feasible sets

- $K_1 = \{x | Ax = b, x \geq 0\}$ set of feasible points for the first stage constrains
- $K_2 = \{x | \mathcal{Q}(x) < +\infty\}$ set of feasible points for the recourse function

These definition allow us to express the determinist equivalent problem in this way:

$$\begin{array}{ll} \min & c^\top x + \mathcal{Q}(x) \\ \text{s. to} & Ax = b \\ & x \geq 0 \end{array} \quad \Longleftrightarrow \quad \begin{array}{ll} \min & c^\top x + \mathcal{Q}(x) \\ \text{s. to} & x \in K_1 \cap K_2 \end{array}$$

Case $\xi(\omega)$ discrete

- We can try defining those x which make a feasible second stage without computing $\mathcal{Q}(x)$.

  We define

  - $K_2(\xi) = \{x | Q(x,\xi) < +\infty\}$ set of points such that $Q(x,\xi)$ is feasible
  - $K_2^P = \{x | \forall \xi \in \Xi \quad \exists y \geq 0 \quad s.t. \quad Wy = h - Tx\} = \bigcap_{\xi \in \Xi} K_2(\xi)$ set of points x such that a feasible second stage decision can be found $\forall \xi \in \Xi$.

Example: $Q(x,\xi) = \begin{cases} \min_{y \in \mathbb{R}^2} & 2y_1 + y_2 \\ s.t.: & y_1 + 2y_2 \geq \xi_1 - x_1 \\ & y_1 + y_2 \geq \xi_2 - x_1 - x_2 \\ & 0 \leq y_1 \leq 1, \quad 0 \leq y_2 \leq 1 \end{cases}, \ \Xi = \left\{\begin{bmatrix} \xi_1 \\ \xi_2 \end{bmatrix}\right\} = \left\{\begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ 7 \end{bmatrix}\right\}$

Case $\xi(\omega)$ discrete

Theorem

- $\forall \xi \in \Xi$, $K_2(\xi)$ is a closed, convex, polyhedral set, and therefore K_2^P is a closed convex set.
- If ξ is finite, K_2^P is polyhedral too and $K_2^P = K_2$.

Case $\xi(\omega)$ general

In this case, we may find strange situations. Example given:

$$\begin{array}{ll} Q(x,\xi) = & \min y \\ \text{s. to} & \xi y = 1 - x \\ & y \geq 0 \end{array}$$

ξ follows a $[0,1]$ triangular distribution

$$f(\xi) = 2\xi$$

In this example the only random element is $W(\omega) = \xi$

- Let's compute K_2^P:
  - $\forall \xi \in (0,1]$, solution of $Q(x,\xi)$ is $y = \frac{1-x}{\xi}$ if $y \geq 0$. Hence, $K_2(\xi) = \{x | x \leq 1\} \quad \forall \xi \in (0,1]$
  - For $\xi = 0$, $\nexists y$ such that $0y = 1 - x$, unless $x = 1$, whereby $K_2(\xi = 0) = \{x | x = 1\}$

  Then, $K_2^P = \{x|x=1\} \bigcap \{x|x \leq 1\} = \{x|x=1\}$

Case $\xi(\omega)$ general

- Compute K_2:

  $\mathcal{Q}(x) = E_\xi[Q(x,\xi)]$. Since the only problematic point ($\xi = 0$) has measure 0, we have that

  $$\mathcal{Q}(x) = \int_0^1 Q(x,\xi)f(\xi)d\xi = \int_0^1 \frac{1-x}{\xi} 2\xi d\xi = 2(1-x)\ \forall x \leq 1$$

  and then $K_2 = \{x | x \leq 1\}$. Observe that $K_2^p \subsetneq K_2$

  - This situation isn't usual when W is fixed, and doesn't happen under some assumptions of ξ.

Case $\xi(\omega)$ general

Theorem

A stochastic program with fixed recourse (W fix) where ξ has finite second moments (i.e., $E(\xi^2) = \int_{\Xi} \xi^2 f(\xi) d\xi < +\infty$) satisfies:

1. K_2 and K_2^P coincide.
2. K_2 is closed and convex.
3. If T is fixed, K_2 is polyhedral.

Outline

Properties of $Q(x, \xi)$, ξ discrete

Theorem

For a stochastic program with fixed recourse, ξ discrete, $Q(x, \xi)$ is

1. a piecewise linear convex function in (h, T).
2. a piecewise linear concave function in q.
3. a piecewise linear convex function in $x \; \forall x \in K_2$

Properties of $\mathcal{Q}(x)$, ξ general

- Now, some results about $\mathcal{Q}(x) = E_\xi[Q(x,\xi)]$

Theorem

For a stochastic program with fixed recourse where ξ has finite second moments

1. $\mathcal{Q}$ is a Lipschitzian convex function and is finite on K_2
   - Finite on K_2: $\mathcal{Q}(x) < +\infty \quad \forall x \in K_2$
   - Lipschitzian: $\exists M \geq 0 : ||\mathcal{Q}(x_2) - \mathcal{Q}(x_1)|| \leq M||x_2 - x_1||\ \forall x_1, x_2 \in K_2$
2. When ξ is finite, $\mathcal{Q}(x)$ is piecewise linear
3. If $F(\xi)$ is abs. continuous (ξ follows a continuous distribution),

$\mathcal{Q}(x)$ is differentiable on $\mathrm{ri}\ K_2$.

Properties of $K_2, K_2^P, Q(x,\xi)$ and $\mathcal{Q}(x)$ ($W(\omega)$ fixed)	**ξ discrete**	**ξ general with $E[\xi^2] < +\infty$**
K_2, K_2^P convex closed.	Always.	Always.
K_2 polyhedral		If $T(\omega)$ fixed.
K_2^P polyhedral.	If ξ finite.	
$K_2 \equiv K_2^P$		Always
$Q(x,\xi)$ piecewise linear and convex in $h, T, x \in K_2$ / concave in q.	Always.	
$\mathcal{Q}(x)$ Lipsch. convex, finite on K_2.	If $E[\xi^2] < +\infty$	
$\mathcal{Q}(x)$ piecewise linear and convex on K_2	If ξ finite.	
$\mathcal{Q}(x)$ convex and differentiable on $\mathrm{ri}\, K_2$.		$F(\xi)$ absolute continuous (ξ continuous distribution).

(Following Birge&Louveaux, 2nd edition)

Recourse function properties in Stochastic Integer Programs

An interesting case is when integer variables appear into second stage decisions. In this case:

Proposition

The expected recourse function $\mathcal{Q}(x)$ of an integer stochastic program is in general non-convex and discontinuous.

As an example, consider a stochastic problem where the first stage contains a single decision variable $x \geq 0$, and the second stage recourse function is defined as:

$$\begin{array}{ll} Q(x,\xi) = & \min 2y_1 + y_2 \\ \text{s. to} & y_1 \geq x - \xi \\ & y_2 \geq \xi - x \\ & y_1, y_2 \in \mathbb{Z}_0^+ \end{array}$$

Recourse function properties in Stochastic Integer Programs

Consider $\xi \in \{1, 2\}$ with equal probability $1/2$.

- Case $\xi = 1$

  If $x < 1 \rightarrow \begin{cases} y_1^* = 0 \\ y_2^* = \lceil 1 - x \rceil = 1 \end{cases}$

  If $x \geq 1 \rightarrow \begin{cases} y_1^* = \lceil x - 1 \rceil \\ y_2^* = 0 \end{cases}$

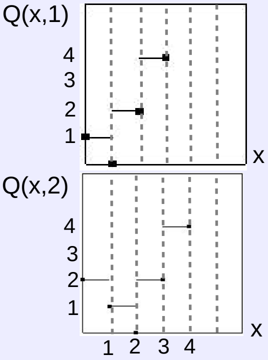
  $Q(x, 1) = \max\{2\lceil x - 1 \rceil, \lceil 1 - x \rceil\}$

- Case $\xi = 2$

  If $x < 2 \rightarrow \begin{cases} y_1^* = 0 \\ y_2^* = \lceil 2 - x \rceil \end{cases}$

  If $x \geq 2 \rightarrow \begin{cases} y_1^* = \lceil x - 2 \rceil \\ y_2^* = 0 \end{cases}$

  $Q(x, 2) = \max\{2\lceil x - 2 \rceil, \lceil 2 - x \rceil\}$



Recourse function properties in Stochastic Integer Programs

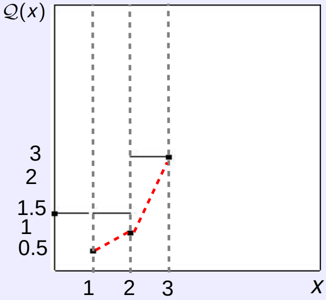
$$Q(x,\xi) = \begin{cases} \lceil 1-x \rceil & 0 \leq x < 1 \\ 2\lceil x-1 \rceil & 1 \leq x \end{cases} \quad \xi = 1$$
$$\qquad\qquad \begin{cases} \lceil 2-x \rceil & 0 \leq x < 2 \\ 2\lceil x-2 \rceil & 2 \leq x \end{cases} \quad \xi = 2$$

Finally, $\mathcal{Q}(x) = 1/2Q(x,1) + 1/2Q(x,2)$

- $\mathcal{Q}(x)$ is discontinuous
- $\mathcal{Q}(x)$ is not convex:

in the points $\begin{array}{l} \mathcal{Q}(1.5) = 1.5 \\ \mathcal{Q}(1) = 0.5 \\ \mathcal{Q}(2) = 1 \end{array}$ it doesn't verify

$$\mathcal{Q}(\lambda x_1 + (1-\lambda)x_2) \leq \lambda \mathcal{Q}(x_1) + (1-\lambda)\mathcal{Q}(x_2)$$



Outline

Definitions

There are 3 kinds of special stochastic problems:

- *Relatively Complete Recourse Problems*

  We say SP has relatively complete recourse if $\forall x \in K_1$, exists a decision y that satisfies the second stage conditions:

  $$\{y \geq 0, Wy = h - Tx\}$$

  In other words, $x \in K_1 \implies x \in K_2$. In set notation, $K_1 \subset K_2$.

- *Complete Recourse Problems*

  Those problems in which $\forall t \in \mathbb{R}^{m_2}$ we can express it in the form

  $$y \geq 0, \quad Wy = t$$

  I.e., $\mathbb{R}^{m_2}$ can be expanded with $\{y \geq 0, Wy\}$. In this case, $K_2 = \mathbb{R}^{n_1}$ and $K_1 \subset K_2$.

Definitions

- *Simple Recourse Problems*

A particular case of a Complete Recourse Problem, where:

$$y = \begin{pmatrix} y^+ \\ y^- \end{pmatrix} \quad W = (\mathbb{I}^+ \quad \mathbb{I}^-) \Longrightarrow \begin{array}{rl} Q(x,\omega) = & \min q^+ y^+ + q^- y^- \\ \text{s. to} & y^+ - y^- = h - Tx \\ & y^+, y^- \geq 0 \end{array}$$

Here, we impose $q_i^+ \geq 0$, $q_i^- \geq 0$ to avoid unbounded solutions, although $q_i^+ + q_i^- \geq 0$ is a weaker necessary condition. Under this assumption the optimal solution is:

$$\begin{cases} y_i^- = 0 \quad y_i^+ = (h_i - T_i x) & \text{if } (h_i - T_i x) \geq 0 \\ y_i^+ = 0 \quad y_i^- = (T_i x - h_i) & \text{if } (h_i - T_i x) \leq 0 \end{cases}$$

Suppose $q_i^+ + q_i^- < 0$ for some i, and consider $y_i^+ + \Delta$, $y_i^- + \Delta$, from a feasible y. $y^+ + \Delta$, $y^- + \Delta$ are feasible if $\Delta \geq 0$ since $(y^+ + \Delta) - (y^- + \Delta) = h - Tx$.

Then, the objective function at coordinate i is $q_i^+(y_i^+ + \Delta) + q_i^-(y_i^- + \Delta) = q_i^+ y_i^+ + q_i^- y_i^- + \Delta(q_i^+ + q_i^-)$. As $(q_i^+ + q_i^-) < 0$, the objective value goes to $-\infty$ as Δ increases.

Simple Recourse Problems

Under $q_i^+ \geq 0$, $q_i^- \geq 0$ assumption, we compute $\mathcal{Q}(x)$:

$$\mathcal{Q}(x) = \sum_{i=1}^{m_2=n_2/2} \mathcal{Q}_i(x) = \sum_{i=1}^{m_2} E_\xi[Q_i(x,\xi)] = \sum_{i=1}^{m_2} E_\xi[q_i^+(h_i - T_i x)^+ + q_i^-(T_i x - h_i)^+]$$

where we have made the notation $(x)^+ = \begin{cases} 0 & \text{if } x < 0 \\ x & \text{if } x \geq 0 \end{cases}$

i.e. $\mathcal{Q}(x) = \sum_{i=1}^{m_2} \mathcal{Q}_i(x)$ is a separable function and

$$\boxed{\mathcal{Q}_i(x) = E_\xi[q_i^+(h_i - T_i x)^+ + q_i^-(T_i x - h_i)^+]}$$

The particular separable structure of the Simple Recourse Problems can be exploited to obtain effective solving methods.

Special SRP Case - only h_i random

For example, consider a case where T_i and q are fixed, the only random component is h_i. Then, we can write and compute $\mathcal{Q}(x)$ in a compact form:

$$\boxed{\mathcal{Q}_i(x) = q_i^+ \bar{h}_i - \left(q_i^+ - q_i F_i(T_i x)\right) T_i x - q_i \int_{h_i \leq T_i x} h_i f_i(h_i) dh_i} \quad (1)$$

where $\bar{h}_i = E[h_i]$ and $q_i = q_i^+ + q_i^-$

(You can check (1) as en exercise)

The News vendor problem

The News vendor problem is a Simple Recourse Problem:

- $x \to$ demand of newspapers to publisher at cost c
- $y \to$ effective sales to clients at price q
- $w \to$ newspapers returned to publisher at price r

$$\begin{array}{ll} \min & c^\top x + E_\xi \left[\begin{array}{ll} \min & -qy - rw \\ \text{s. to} & y \le \xi \\ & y + w \le x \\ & y, w \ge 0 \end{array} \right] \\ \text{s. to} & 0 \le x \le u \end{array}$$

$$\left\{ \begin{array}{ll} \text{if } x \le \xi & \left\{ \begin{array}{l} y = x \\ w = 0 \end{array} \right. \\ \text{if } x \ge \xi & \left\{ \begin{array}{l} y = \xi \\ w = x - \xi \end{array} \right. \end{array} \right.$$

The News vendor problem

It can be showed that $Q(x,\xi)$ can be written as a Simple Recourse Problem in the following way:

- Use the equation $x+y^+ - y^- = \xi$ where y^+ and y^- are the s lack and surplus of newspapers, respectively.
- Now, we have $\begin{cases} w = y^- \\ y = \xi - y^+ \end{cases}$

We can write $Q(x,\xi)$ as

$$\begin{array}{ll} \min & -qy - rw \\ \text{s. to} & x + y^+ - y^- = \xi \\ & y^+, y^- \geq 0 \end{array} \Rightarrow \begin{array}{ll} \min & -q(\xi - y^+) - r(y^-) \\ \text{s. to} & y^+ - y^- = \xi - x \\ & y^+, y^- \geq 0 \end{array} \Rightarrow \begin{array}{ll} -q\xi + \min & qy^+ - ry^- \\ \text{s. to} & y^+ - y^- = \xi - x \\ & y^+, y^- \geq 0 \end{array}$$

that verifies $q - r > 0$ and therefore solution is:

$$\begin{cases} \text{if } \xi - x \geq 0 & \begin{cases} y^+ = \xi - x & \rightarrow y = \xi - (\xi - x) = x \\ y^- = 0 & \rightarrow w = 0 \end{cases} \\ \text{if } \xi - x < 0 & \begin{cases} y^+ = 0 & \rightarrow y = \xi - 0 = \xi \\ y^- = x - \xi & \rightarrow w = x - \xi \end{cases} \end{cases}$$

That coincides with original solution of the problem formulated with variables y and w.

The News Vendor Problem

Therefore,

$$\mathcal{Q}(x) = E_\xi[-q\xi + q(\xi - x)^+ - r(x-\xi)^+] = -qE[\xi] + E_\xi[q(\xi - x)^+ - r(x-\xi)^+] = -qE[\xi] + A$$

and problem has the form

$$\begin{array}{ll} \min & c^\top x + \mathcal{Q}(x) \\ \text{s. to} & 0 \leq x \leq u \end{array}$$

We can define the objective function ever further, because the only random element $h = \xi$, with density function F.

$$A = E_\xi[q(\xi - x)^+ - r(x-\xi)^+] = \left| (1) \begin{cases} q^+ \equiv q & T \equiv I \\ q^- \equiv r & h_i \equiv \xi \end{cases} \bar{\xi} \equiv E[\xi] \right| = q\bar{\xi} - \big(q - (q-r)F(x)\big)x - (q-r)\int_{-\infty}^{x} \xi f(\xi) d\xi$$

$$\mathcal{Q}(x) = -(q-(q-r)F(x))x - (q-r)\int_{-\infty}^{x} \xi f(\xi) d\xi = -qx + (q-r)\left[F(x)x - \int_{-\infty}^{x} \xi f(\xi) d\xi\right]$$